

PlenoGS: Direct 3D Gaussian Splatting Training from Plenoptic Camera Images

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Introduction

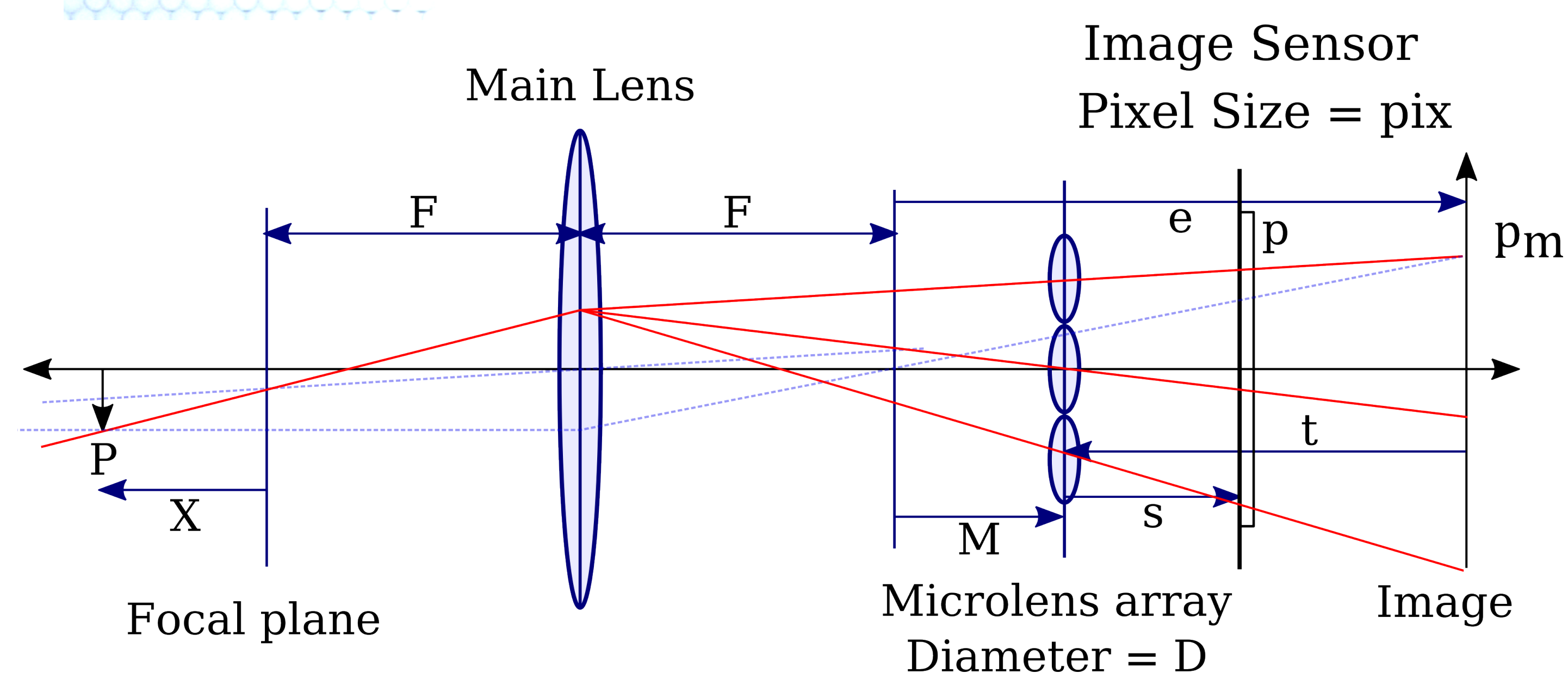
We train 3D Gaussian Splatting directly on plenoptic images without requiring subaperture view extraction. To achieve this, we propose a differentiable framework implemented in *gsplat* for projecting and rasterizing 3D Gaussians through a plenoptic camera.

Plenoptic Camera



$$p_m = K_m \times T_m \times K_M \times P$$

pixel Micro-lens Main lens Main lens 3D point
to
Micro-lens

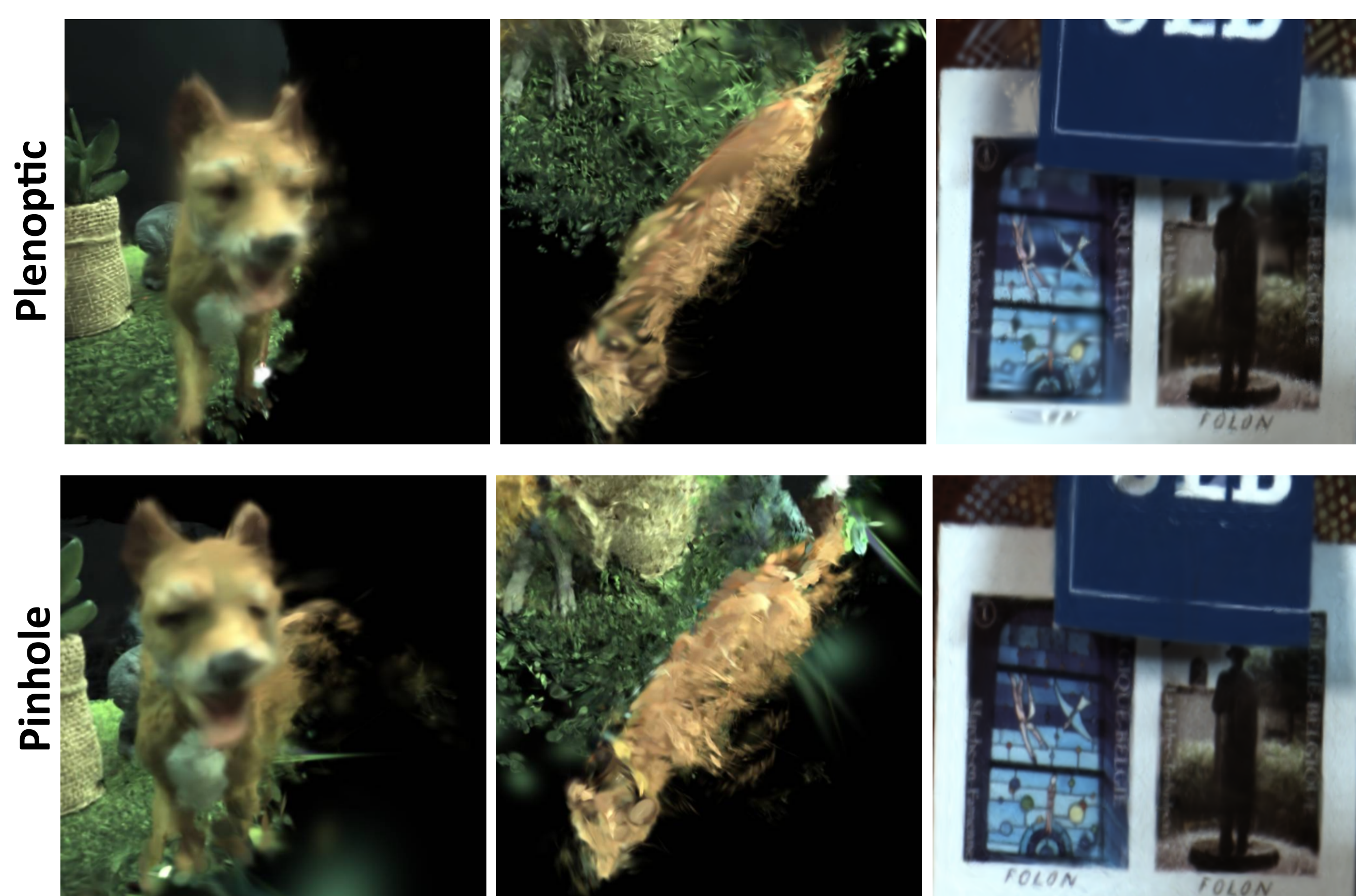


Results

Dataset	psnr	ssim	lpips	Time	#Gaussians
Fujita	21.17	0.61	0.14	5m 36s	49,555
Rabbit	23.95	0.65	0.14	10m 33s	55,591
Temple Boat (1k)	21.25	0.59	0.28	9m 48s	94,063
Cubes	14.49	0.162	0.407	5m 07s	38,573

Datasets have been trained for 3000 iterations except for Temple Boat.

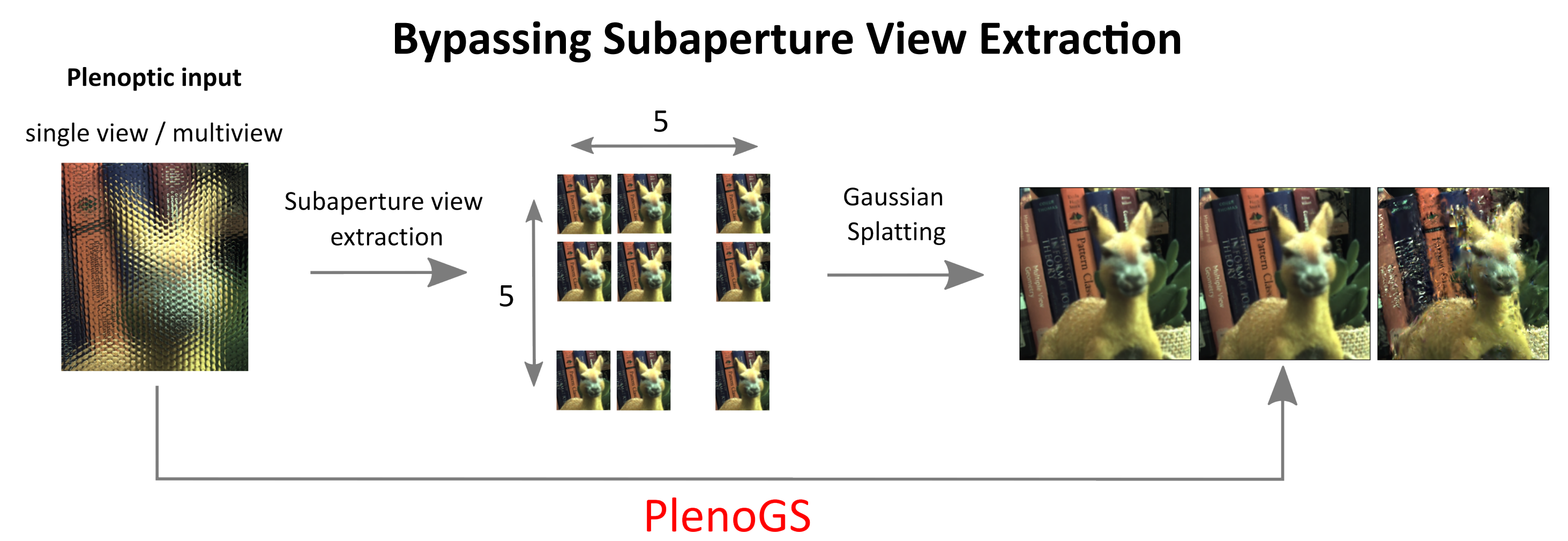
Plenoptic vs. Pinhole Training



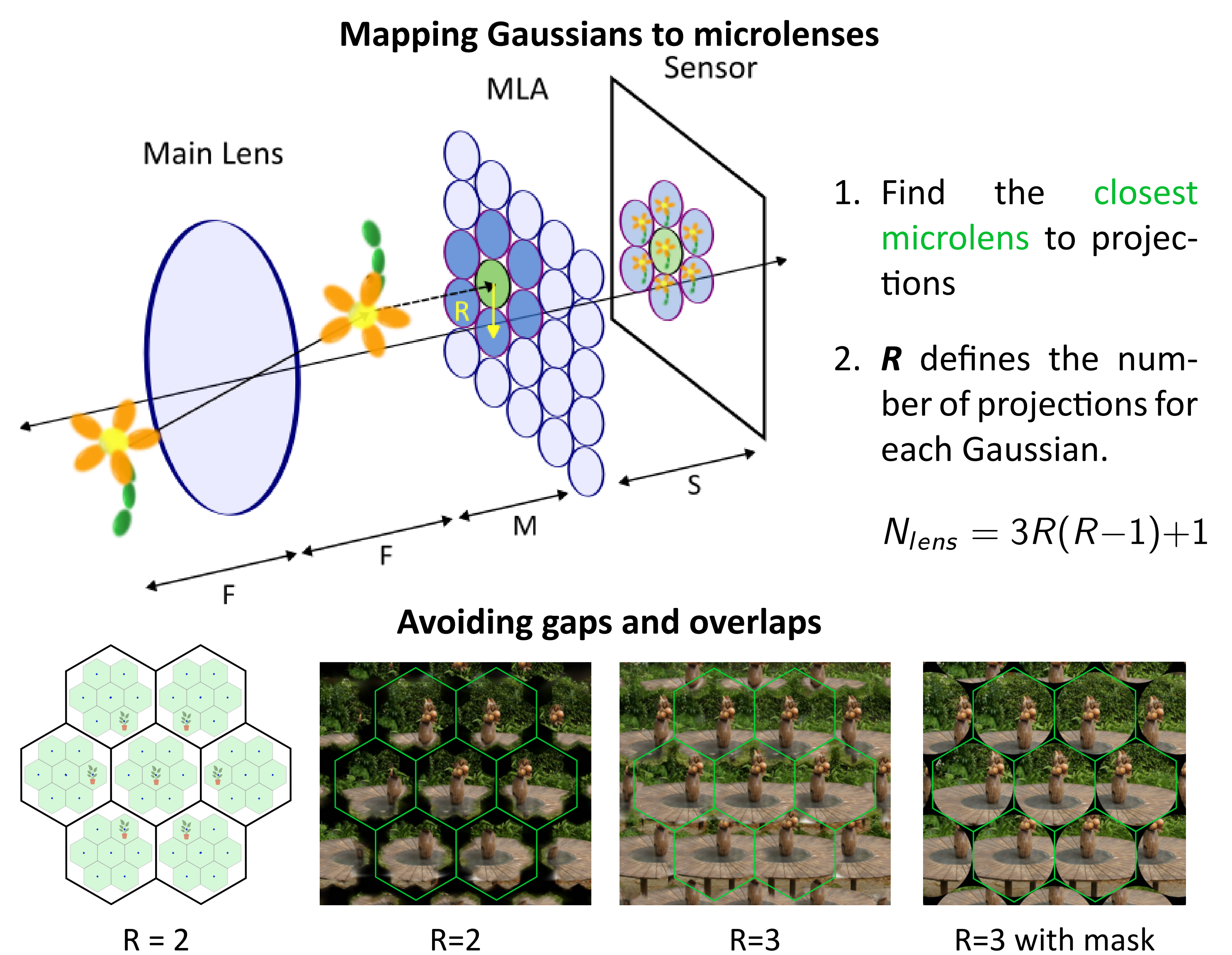
Acknowledgment

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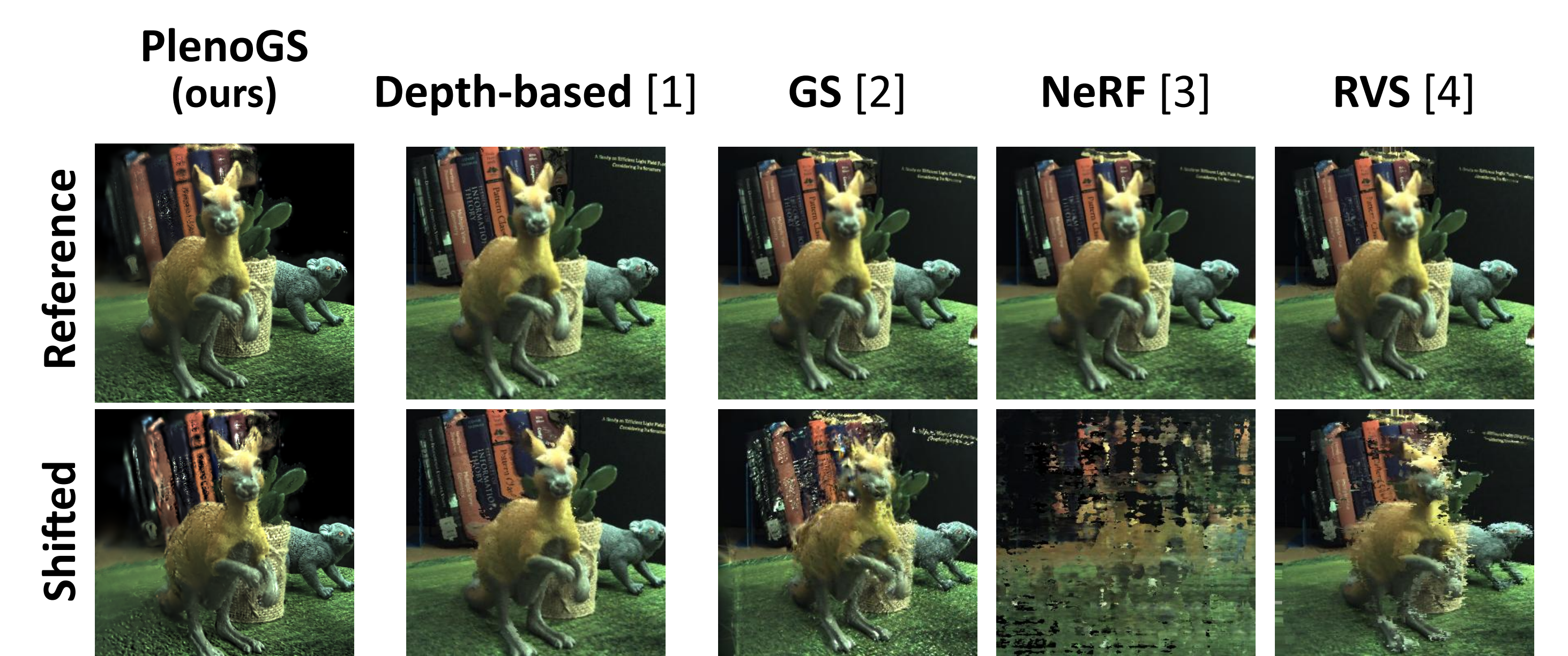
Goal



Method



PlenoGS vs. Existing Methods



- [1] Fachada et al. "Micro-Image Domain View Synthesizer for Free Navigation With Focused Plenoptic Cameras," IEEE Transactions on Multimedia, vol. 27, pp. 7179–7191, 2025.
 [2] Kerbl et al. "3D Gaussian Splatting for Real-Time Radiance Field Rendering," ACM Transactions on Graphics, vol. 42, pp. 1–14, Aug. 2023.
 [3] Mildenhall et al. "NeRF: representing scenes as neural radiance fields for view synthesis," Commun. ACM, vol. 65, no. 1, pp. 99–106, 2021.
 [4] Bonatto et al. "Real-time depth video-based rendering for 6-dof hmd navigation and light field displays," IEEE Access, vol. 9, pp. 146868–146887, 2021.

Conclusion

Our method is the first to train Gaussian Splatting directly on plenoptic images, enhancing 3D reconstruction compared to pipelines requiring subaperture extraction.

Future Work

Future work includes depth-supervised loss, adaptive R based on scene depth and modeling microlenses as multifocal thin lenses.